UC SAN DIEGO / ECE 251C

14
Finite wavelet transforms and signal recovery

Boundary handling, computational cost, image wavelets, and inverse problems

How many wavelet coefficients do we need to reconstruct an N-point signal?

14.1 Computational and numerical considerations

Recall that the DWT is a low-pass-iterated analysis bank of a perfect-reconstruction filter bank. At each stage, the input is decomposed into approximation and detail coefficients. The approximation coefficients describe the signal at a lower resolution, while the detail coefficients record the information needed to recover the finer resolution. We retain the details and repeat the decomposition on the approximation branch.

At resolution $i+1$, the analysis equations are

$$a_i[n] = (h_0 * a_{i+1})[2n], \qquad d_i[n] = (h_1 * a_{i+1})[2n]. \tag{14.1}$$

In practice, we work with a finite number of samples. Suppose the input is an N-point signal. How many coefficients do the two output sequences have? Although each branch is downsampled by two, convolution may first increase its length. We therefore need to examine the filtering step and specify how the signal is extended beyond its endpoints.

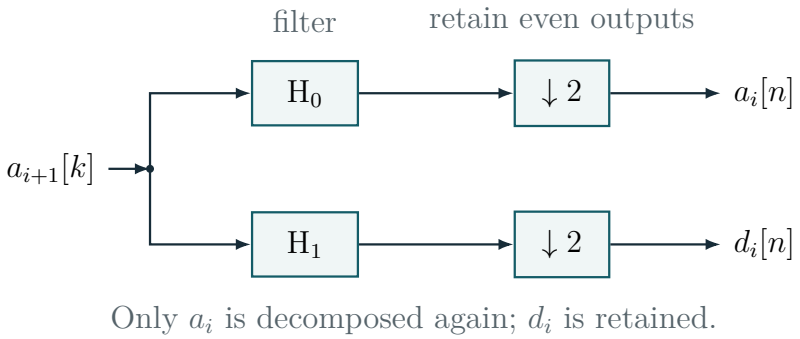


Figure 14.1: One stage of the DWT. Filtering produces the approximation and detail branches, and downsampling retains their even-indexed outputs. Only the approximation branch is decomposed again.

14.2 Length of a convolution

An N-point signal has N stored samples, some of which may be zero. Suppose $x[n]$ is supported in $\{0, \dots, N-1\}$ and $h[n]$ in $\{0, \dots, L-1\}$. Extend both sequences by zero. Their convolution is

$$y[k] = (h * x)[k] = \sum_{r=0}^{L-1} h[r]x[k-r].$$

A nonzero summand requires both $0 \le r \le L-1$ and $0 \le k-r \le N-1$. Therefore

$$0 \le k \le N + L - 2.$$

Thus the convolution has length at most $N + L - 1$. When the first and last entries of both sequences are nonzero,

$$y[0] = h[0]x[0] \neq 0, \qquad y[N+L-2] = h[L-1]x[N-1] \neq 0,$$

and the length is exactly $N + L - 1$. Some entries between these endpoints may be zero.

Keeping even indices $k = 2n$ from that full array gives

$$0 \le n \le \left\lfloor \frac{N+L-2}{2} \right\rfloor.$$

Hence the number of retained positions *per branch* is

$$\boxed{\left\lfloor \frac{N+L-2}{2} \right\rfloor + 1 = \left\lceil \frac{N+L-1}{2} \right\rceil}. \tag{14.2}$$

For even N and even L, this becomes $(N + L)/2$. Computing both full branches then produces $N + L$ positions, rather than N.

Observe that the two downsampled outputs may contain more coefficients than the original signal. This increase comes from the convolution near the boundaries. Since all output coefficients are determined by the N input samples, they cannot be independent. This does not, however, tell us which coefficients can be discarded. To understand the redundancy, we examine the Haar case explicitly.

14.3 Example: a finite-length Haar transform

For this boundary calculation, use the *causal* analysis filters

$$h_0^{\mathrm{c}}[k] = \frac{\delta[k] + \delta[k-1]}{\sqrt{2}}, \qquad h_1^{\mathrm{c}}[k] = \frac{\delta[k] - \delta[k-1]}{\sqrt{2}}.$$

Consider the six-point signal $x[0] = 1$, $x[1] = 3$, $x[2] = x[3] = 2$, and $x[4] = x[5] = 4$, extended by zero. Each convolution has seven entries. Downsampling the full output at even indices gives four approximation coefficients and four detail coefficients.

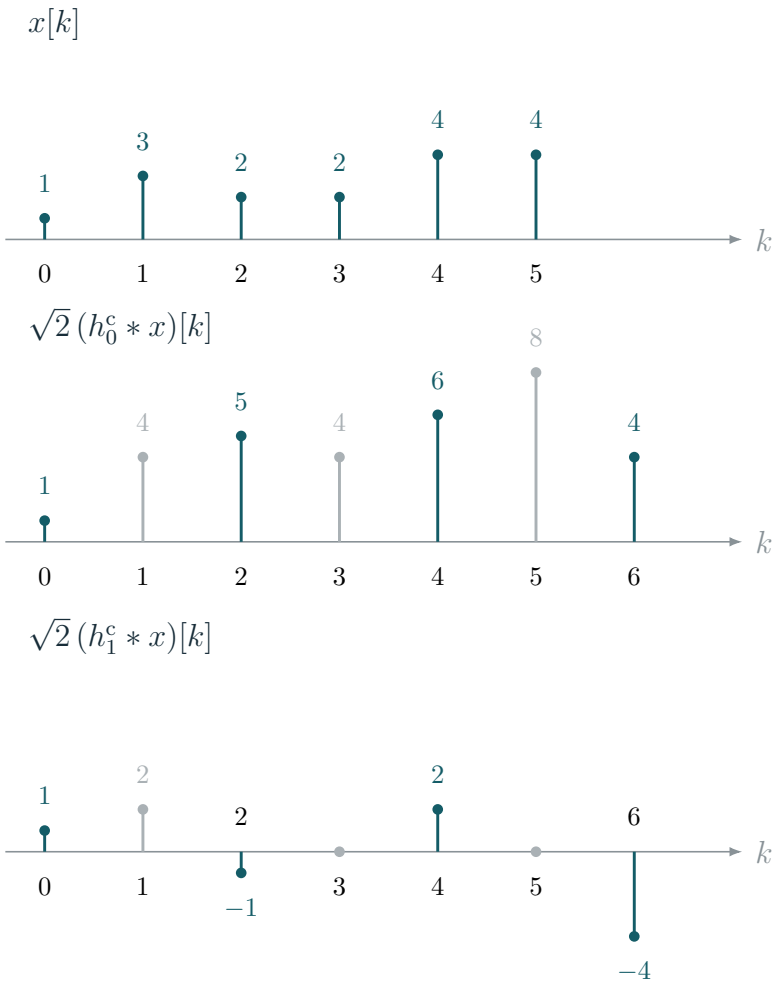


Figure 14.2: A six-sample example with causal Haar filters. Filtered values are shown after multiplication by $\sqrt{2}$. In the filtered sequences, teal marks the retained even outputs and gray marks the discarded odd outputs. The last even outputs contain the last input sample.

Which coefficients are redundant? Write the six input entries as $x_0, \ldots, x_5$. With causal filters and even-output retention,

$$a^{\mathrm{c}}[n] = \frac{x[2n] + x[2n-1]}{\sqrt{2}}, \qquad d^{\mathrm{c}}[n] = \frac{x[2n] - x[2n-1]}{\sqrt{2}}.$$

The four output pairs are

n	$\sqrt{2}\, a^{\mathrm{c}}[n]$	$\sqrt{2}\, d^{\mathrm{c}}[n]$
0	x_0	x_0
1	$x_2 + x_1$	$x_2 - x_1$
2	$x_4 + x_3$	$x_4 - x_3$
3	x_5	$-x_5$

Thus $d^{\mathrm{c}}[0] = a^{\mathrm{c}}[0]$ and $d^{\mathrm{c}}[3] = -a^{\mathrm{c}}[3]$. One may retain all four approximation coefficients and only the two interior detail coefficients. They recover the input by

$$x_0 = \sqrt{2}\, a^{\mathrm{c}}[0], \qquad x_5 = \sqrt{2}\, a^{\mathrm{c}}[3],$$

$$x_{2n} = \frac{a^{\mathrm{c}}[n] + d^{\mathrm{c}}[n]}{\sqrt{2}}, \qquad x_{2n-1} = \frac{a^{\mathrm{c}}[n] - d^{\mathrm{c}}[n]}{\sqrt{2}}, \quad n = 1, 2.$$

Thus six coefficients suffice to recover the six input samples. The particular coefficients retained depend on the alignment: in this causal convention, both final outputs contain x_5, and discarding both would lose that sample.

The adjacent-pair convention used earlier. The zero-delay analysis filters $h_\ell[k] = g_\ell[-k]$ for $g_0 = (1, 1)/\sqrt{2}$ and $g_1 = (1, -1)/\sqrt{2}$ give

$$a[n] = \frac{x[2n] + x[2n+1]}{\sqrt{2}}, \qquad d[n] = \frac{x[2n] - x[2n+1]}{\sqrt{2}}. \tag{14.3}$$

Here the samples are grouped into the pairs $(x[0], x[1])$, $(x[2], x[3])$, and $(x[4], x[5])$. Consequently, $n = 0, 1, 2$ gives three approximation coefficients and three detail coefficients. This is the zero-delay convention from the earlier lectures, which we use below.

14.4 The periodic DWT

For longer filters, linear convolution introduces more boundary coefficients. Rather than treating these separately, we can change the boundary convention. In a *periodic DWT*, the finite signal is extended periodically and each filtering operation is a circular convolution. This keeps the total number of approximation and detail coefficients equal to the signal length.

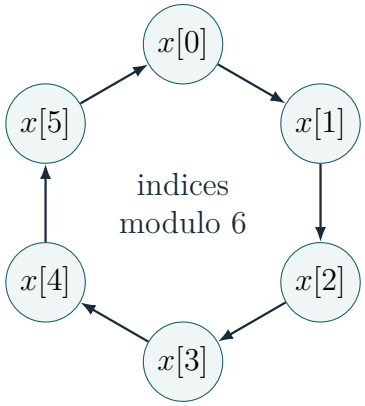
For an N-point signal, define

$$x_{\mathrm{per}}[k] = x[k \bmod N], \qquad k \in \mathbb{Z},$$

where $k \bmod N$ is the representative in $\{0, \dots, N-1\}$. For a finitely supported filter h, define

$$(h \circledast_N x)[k] = \sum_{r \in \operatorname{supp} h} h[r]x[(k-r) \bmod N], \qquad 0 \le k < N. \tag{14.4}$$

The modulo operation replaces an index outside the stored range by the corresponding index in one period. In particular, $x[-1]$ is interpreted as $x[N-1]$. The sum includes all filter taps, even when the filter is longer than the current coefficient sequence.



$x[-1] = x[5]$

$x[6] = x[0]$

$x[k+6] = x[k]$

The record length stays six. Wrapping changes its boundary model.

Figure 14.3: Circular convolution treats the endpoints as neighbors. Only one period is stored; sample values outside this period are obtained by wrapping their indices.

For even N, the periodic analysis stage is

$$\begin{aligned} a[n] &= (h_0 \circledast_N x)[2n], \\ d[n] &= (h_1 \circledast_N x)[2n], \end{aligned} \qquad 0 \le n < N/2. \tag{14.5}$$

There are $N/2$ coefficients in each branch, so the total remains N. This is analogous to the periodic extension implicit in the DFT. The synthesis bank must use the same convention. A perfect-reconstruction delay then becomes a circular shift, which can be compensated to recover the original alignment.

14.5 Boundary effects

Periodic extension treats the last and first samples as neighbors. If those values do not match, it introduces an artificial jump. Wavelets near this boundary respond to the jump just as they respond to a transition within the signal.

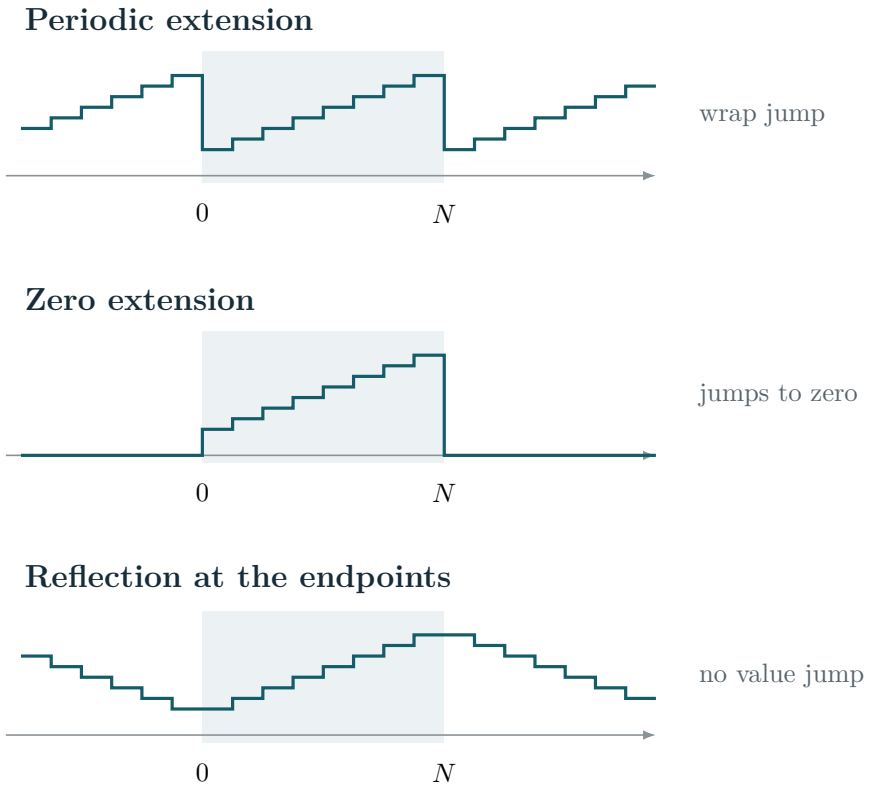


Figure 14.4: The shaded segment is the stored record. Periodization, zero extension, and reflection give different continuations beyond its endpoints. These continuations affect the wavelet coefficients whose filters overlap the boundary.

Zero padding places the signal inside a longer sequence of zeros, so the periodic wrap can occur in a zero region. It does not remove the transitions from nonzero endpoints to zero. For a single convolution of lengths N and L, padding to at least $N + L - 1$ prevents wrap-around of the full output.

Another option is to reflect the signal at its endpoints. This gives a different boundary convention. We use the periodic DWT for the coefficient counts and computational costs below.

These boundary effects do not prevent perfect reconstruction when the analysis and synthesis conventions agree. They become important when we modify the coefficients: thresholding may alter a coefficient associated with an artificial boundary rather than a feature of the original signal.

14.6 Computational complexity

Suppose $h[n]$ has length L. Each output of the circular convolution is a sum of L products. Direct evaluation therefore requires L multiplications and $L-1$ additions per output, or NL multiplications and $N(L-1)$ additions for all N outputs.

For a decimated output, compute only

$$(h \circledast_N x)[2n] = \sum_{r \in \operatorname{supp} h} h[r]x[(2n-r) \bmod N], \qquad 0 \le n < N/2.$$

There are only $N/2$ retained outputs, so direct evaluation requires $NL/2$ multiplications and $N(L-1)/2$ additions. The saving comes from computing only the outputs that will be used. Computing the full convolution and then discarding the odd-indexed outputs would perform unnecessary work.

For the DWT, suppose both analysis filters have length L and $N = 2^I$. The first stage computes both a_{I-1} and d_{I-1}. Each has $N/2$ entries, so the total is LN multiplications. Continuing along the low-pass branch gives the following counts.

Input	Length	Outputs	Each length	Multiplications
a_I	N	a_{I-1}, d_{I-1}	$N/2$	LN
a_{I-1}	$N/2$	a_{I-2}, d_{I-2}	$N/4$	$LN/2$
a_{I-2}	$N/4$	a_{I-3}, d_{I-3}	$N/8$	$LN/4$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
a_{i+1}	2^{i+1}	a_i, d_i	2^i	$L2^{i+1}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
a_1	2	a_0, d_0	1	$2L$

At resolution i, both a_i and d_i have 2^i entries. Equivalently, after r stages, a_{I-r} and d_{I-r} each have $N/2^r$ entries.

14.7 Total cost of the DWT

At each stage, we retain the details and continue only with the approximation coefficients. The input length is therefore halved at every stage. Summing the multiplication counts gives

$$\begin{aligned} C_{\text{mult}} &= LN + \frac{LN}{2} + \frac{LN}{4} + \cdots + 2L \\ &= L \sum_{i=0}^{I-1} 2^{i+1} = 2L(2^I - 1) = \boxed{2L(N-1)}. \end{aligned} \tag{14.6}$$

Likewise,

$$C_{\text{add}} = \boxed{2(L-1)(N-1)}. \tag{14.7}$$

These counts are for direct evaluation of the two filters. Special filter coefficients may allow fewer operations. If the filter lengths are L_0 and L_1, the multiplication count is $(L_0 + L_1)(N-1)$.

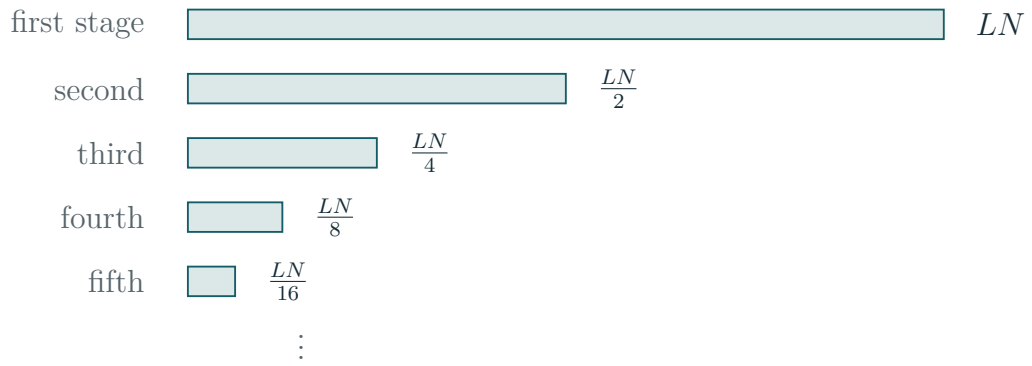


Figure 14.5: Work per stage in direct evaluation of both branches. Each stage has half as many input coefficients as the preceding one. The sum is less than twice the first-stage cost.

Therefore, the total computational cost is $O(LN)$. For a fixed pair of filters, L is independent of N, and the cost is $O(N)$. In particular, short-tap filters give a fast DWT.

The standard radix-two FFT costs $O(N \log N)$. For the DWT, the work over all resolutions forms a geometric sum: the second stage costs half as much as the first, the third costs one-quarter as much, and so on. Equivalently, its cost satisfies

$$T_{\text{DWT}}(N) = T_{\text{DWT}}(N/2) + O(LN).$$

Thus the presence of $I = \log_2 N$ stages does not introduce a logarithmic factor. The total work is less than twice the first-stage cost.

14.8 The total number of coefficients

After the full decomposition, we keep the coarsest approximation a_0 and all the detail sequences $d_0, \ldots, d_{I-1}$. One ordering of the output coefficients is

$$(a_0[0],\ d_0[0],\ d_1[0], d_1[1],\ \ldots,\ d_{I-1}[0], \ldots, d_{I-1}[2^{I-1}-1]). \tag{14.8}$$

There are exactly

$$\underbrace{1}_{a_0} + \underbrace{\sum_{i=0}^{I-1} 2^i}_{\text{all details}} = 1 + (2^I - 1) = N$$

entries. For $N = 8$, the three stages are

$$\begin{aligned} a_3\ (8) &\longrightarrow a_2\ (4),\ d_2\ (4), \\ a_2\ (4) &\longrightarrow a_1\ (2),\ d_1\ (2), \\ a_1\ (2) &\longrightarrow a_0\ (1),\ d_0\ (1). \end{aligned}$$

The retained lengths are $1 + 1 + 2 + 4 = 8$, while the direct multiplication count is $8L + 4L + 2L = 14L$.

Observe that the DWT itself does not compress the signal: it replaces N samples by N coefficients. Its value is that these coordinates may have very different magnitudes. When most coefficients are small, retaining only the larger ones can give an accurate approximation. Quantizing and encoding the retained coefficients then yields a compressed representation.

The intermediate approximations $a_1, \ldots, a_{I-1}$ need not be stored as additional outputs. Each is decomposed into the next approximation and detail sequence. The required storage is therefore $O(N)$.

For reconstruction, combine a_0 and d_0 to recover a_1, then combine a_1 and d_1 to recover a_2, and continue. The inverse DWT also costs $O(LN)$.

14.9 Two-dimensional DWTs

The same construction extends to images by applying the one-dimensional analysis bank along each coordinate. First transform every row. Each resulting array is then transformed along its columns. This is called a *separable* two-dimensional DWT because it is implemented using one-dimensional operations. All convolutions in this section are circular.

Let $X[r, c]$ be a $(2N) \times (2N)$ image, where r is the row index and c is the column index. Filtering each row and downsampling along the column index gives

$$U_\ell[r, m] = \sum_k h_\ell[k] X[r, (2m - k) \bmod 2N], \qquad \ell = 0, 1.$$

Apply the analysis bank to each column of these two arrays:

$$C_{\ell s}[n, m] = \sum_q h_s[q] U_\ell[(2n - q) \bmod 2N, m]. \tag{14.9}$$

Here $0 \le n < N$ and $0 \le m < N$. Substitution gives the separable formula

$$C_{\ell s}[n, m] = \sum_{q,k} h_s[q] h_\ell[k] X[(2n - q) \bmod 2N, (2m - k) \bmod 2N].$$

The four outputs are $\mathrm{LL} = C_{00}$, $\mathrm{LH} = C_{01}$, $\mathrm{HL} = C_{10}$, and $\mathrm{HH} = C_{11}$. The first letter indicates the filter applied along each row; the second indicates the filter applied along each column. Each output is an $N \times N$ array.

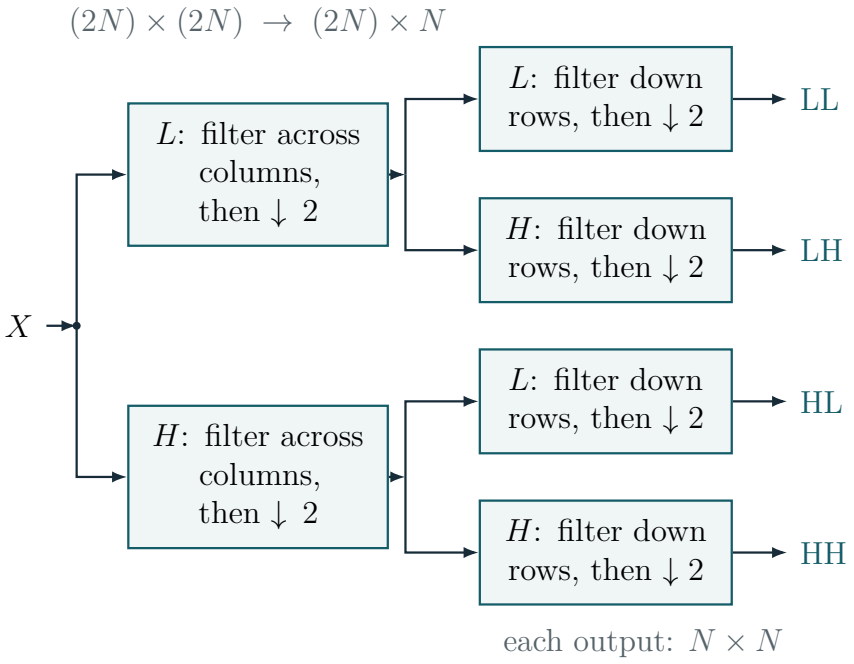


Figure 14.6: A separable two-dimensional analysis bank. The first letter specifies the horizontal filter and the second specifies the vertical filter. Each branch contains an $N \times N$ array, so the total coefficient count remains $(2N)^2$.

14.10 Example: the two-dimensional Haar transform

Use the adjacent-pair Haar convention in (14.3). For a pixel block

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

the horizontal low-pass values are $(a+b)/\sqrt{2}$ and $(c+d)/\sqrt{2}$, and the horizontal high-pass values are $(a-b)/\sqrt{2}$ and $(c-d)/\sqrt{2}$. A second average–difference operation yields

$$\boxed{\begin{aligned} \mathrm{LL} &= (a+b+c+d)/2, \quad \mathrm{LH} = (a+b-c-d)/2, \\ \mathrm{HL} &= (a-b+c-d)/2, \quad \mathrm{HH} = (a-b-c+d)/2. \end{aligned}} \tag{14.10}$$

Each one-dimensional stage contributes a factor $1/\sqrt{2}$, giving the factor $1/2$. The LL coefficient describes the local average, up to normalization. The other three coefficients describe differences across rows, across columns, or in both directions.

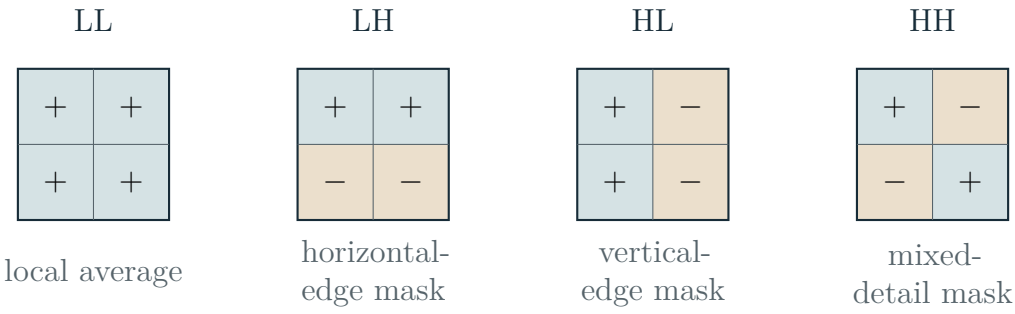


Figure 14.7: The four Haar masks on a 2×2 pixel block. Each sign represents $\pm 1/2$, and the masks are orthonormal. The detail masks distinguish changes across rows, across columns, and in both directions.

For example, $\left[\begin{smallmatrix} 1 & 3 \\ 1 & 3 \end{smallmatrix}\right]$ gives

$$(\mathrm{LL}, \mathrm{LH}, \mathrm{HL}, \mathrm{HH}) = (4, 0, -2, 0).$$

The left-to-right change appears in HL; a top-to-bottom change appears in LH. A diagonal edge can contribute to several bands.

Adding and subtracting the four equations gives the inverse:

$$\begin{aligned} a &= (\mathrm{LL}+\mathrm{LH}+\mathrm{HL}+\mathrm{HH})/2, \quad b = (\mathrm{LL}+\mathrm{LH}-\mathrm{HL}-\mathrm{HH})/2, \\ c &= (\mathrm{LL}-\mathrm{LH}+\mathrm{HL}-\mathrm{HH})/2, \quad d = (\mathrm{LL}-\mathrm{LH}-\mathrm{HL}+\mathrm{HH})/2. \end{aligned}$$

The four masks are orthonormal, so

$$a^2 + b^2 + c^2 + d^2 = \mathrm{LL}^2 + \mathrm{LH}^2 + \mathrm{HL}^2 + \mathrm{HH}^2.$$

14.11 Multilevel image decompositions

One stage maps a $(2N) \times (2N)$ image into four $N \times N$ arrays. The total coefficient count is unchanged:

$$4N^2 = (2N)^2.$$

The LL array approximates the image at a lower resolution. Apply the analysis bank again to this array and retain the other three arrays as details. Repeating this procedure gives a multiresolution decomposition, as in the one-dimensional DWT.

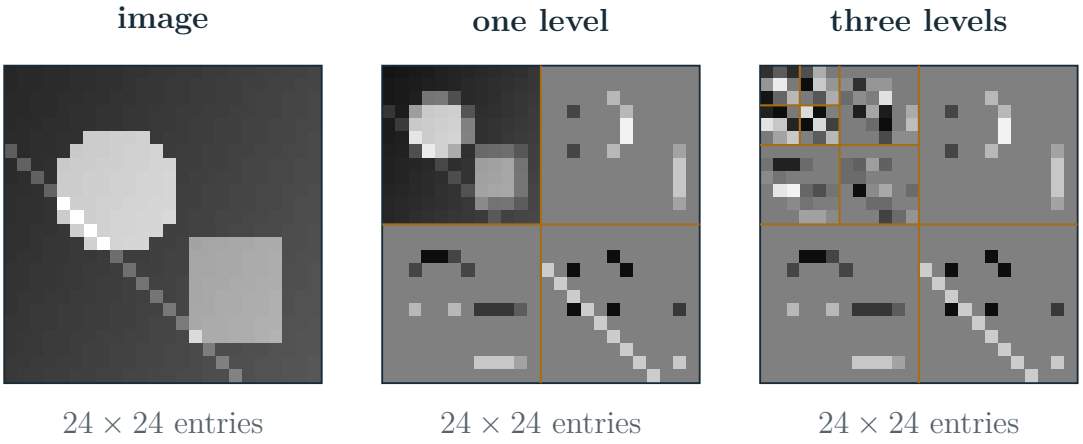


Figure 14.8: A computed Haar decomposition of a 24×24 image. The top row contains LL, HL and the bottom row LH, HH. Only the upper-left block is decomposed again. Each detail band uses its own display scale, with zero in midgray.

For a $2^I \times 2^I$ image, the final approximation has one coefficient, and resolution i has three detail arrays with 4^i entries each. Thus

$$1 + 3 \sum_{i=0}^{I-1} 4^i = 4^I.$$

For the displayed 24×24 example, three stages leave a 3×3 approximation, with $9 + 3(9 + 36 + 144) = 576$ coefficients in total.

Cost per pixel. For a $(2N) \times (2N)$ input, the horizontal passes cost $4LN^2$ multiplications and the vertical passes cost another $4LN^2$. The next LL block has one-quarter as many entries. For a square image of side $S = 2^I$, the full count is

$$2LS^2 \sum_{r=0}^{I-1} 4^{-r} = \frac{8}{3} L(S^2 - 1) = O(LS^2). \tag{14.11}$$

For fixed filters, the cost is linear in the number of pixels, not in the side length. Reconstruction reverses the vertical and horizontal synthesis steps.

14.12 Nonlinear processing of wavelet coefficients

Recall that the wavelet coefficients of a piecewise-smooth signal are often zero or small away from its transitions. This motivates a simple approach to compression and denoising: compute the DWT, modify the coefficients, and reconstruct. The representation is useful because a small number of coefficients can account for much of the signal's energy.

The modification step is essential. Applying the DWT and then its inverse returns the original signal, including any noise. To obtain a different estimate, we must change the coefficients before applying the synthesis bank.

For example, *hard thresholding* sets small coefficients to zero and leaves the remaining coefficients unchanged. For a threshold $\tau > 0$, define

$$\eta_\tau(z) = \begin{cases} 0, & |z| \le \tau, \\ z, & |z| > \tau. \end{cases}$$

This rule is nonlinear. Indeed, $\eta_\tau(3\tau/4) = 0$, whereas $\eta_\tau(3\tau/2) = 3\tau/2$. Thus applying the rule to a sum need not give the sum of the separately thresholded coefficients.

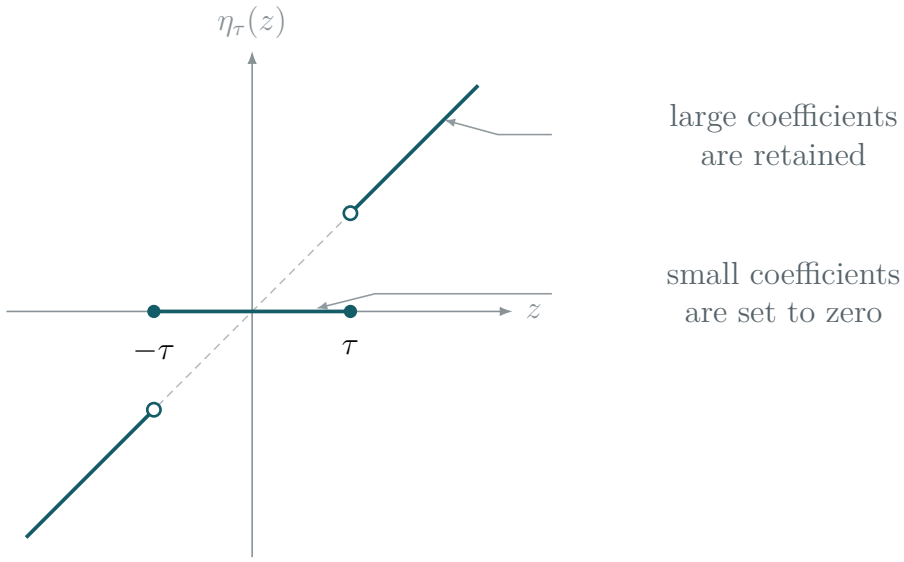


Figure 14.9: Hard thresholding with the convention $\eta_\tau(z) = 0$ at $|z| = \tau$. The dashed diagonal is the identity map. The thresholding rule is nonlinear and discontinuous; the DWT itself is linear.

In a noisy signal, coefficients that were zero need not remain zero. Thresholding can suppress these perturbations, but it may also remove small signal coefficients. The threshold therefore affects both noise removal and the preservation of signal detail.

14.13 Denoising, estimation, and recovery of signals

Suppose we observe

$$\boxed{y[n] = x[n] + \varepsilon[n],} \tag{14.12}$$

where $x[n]$ is the signal of interest, $\varepsilon[n]$ is additive noise, and $y[n]$ is the observed signal. Our goal is to estimate $x[n]$ from $y[n]$. This problem is called *denoising*. Wavelet thresholding is one approach: it uses the structure of the signal's wavelet coefficients to distinguish the signal from the noise.

More generally, we often observe $x[n]$ through a channel, an acquisition device, or a sampling system. Denote this system by H. If noise is added to its output, the observation model is

$$y[n] = (\mathrm{H}x)[n] + \varepsilon[n].$$

The problem is now to estimate the input of the system from its noisy output.

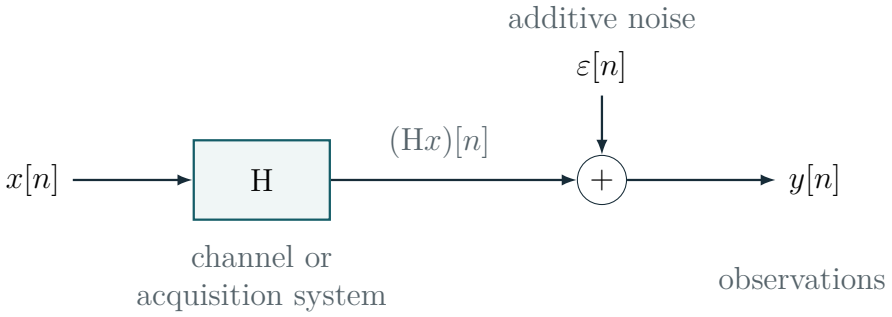


Figure 14.10: The sequence-domain observation model. The system transforms the signal, then additive noise corrupts its output. We seek to estimate the input from these observations.

The system H may or may not be time invariant. In many applications, however, it is linear. This means that we can describe the acquisition process using linear algebra, even when it cannot be described by convolution with a single impulse response.

We have so far studied how to represent signals and reconstruct them from their coefficients. Signal recovery introduces a further question: how should we use the observations, together with prior knowledge about the signal, to construct an estimate? This leads to systems of equations, least squares, and regularization.

14.14 From a finite sequence to a vector

For an N-point real signal, collect the samples into a column vector:

$$x = \begin{pmatrix} x[0] \\ x[1] \\ \vdots \\ x[N-1] \end{pmatrix} \in \mathbb{R}^N.$$

The entries of this vector are the same signal samples as before. The change in notation lets us describe a linear acquisition system by a matrix. If the system produces M measurements, its matrix has M rows and N columns, and the observation model becomes

$$\boxed{y = \mathrm{H}x + \varepsilon,} \tag{14.13}$$

where

$y \in \mathbb{R}^M$	observations
$\mathrm{H} \in \mathbb{R}^{M \times N}$	system matrix
$x \in \mathbb{R}^N$	signal of interest
$\varepsilon \in \mathbb{R}^M$	additive noise

In particular, each row of H specifies one measurement:

$$y[m] = \sum_{n=0}^{N-1} H_{mn} x[n] + \varepsilon[m], \qquad m = 0, \ldots, M-1.$$

The noise has the same dimension as the observations, since these vectors are added. The number of measurements M need not equal the number of unknown signal samples N.

> **Signal recovery as an inverse problem**
> Given the observations y and the system matrix H, estimate the signal x.

Denoising is the special case $M = N$ and $\mathrm{H} = \mathrm{I}_N$. More generally, the matrix need not be square or invertible. The phrase "inverse problem" refers to recovering the input from the output; it does not assume that an ordinary matrix inverse exists.

14.15 Least squares and regularization

If there is no noise, the observations satisfy

$$y = \mathrm{H}x.$$

We therefore seek a signal consistent with these equations. When noise is present, exact consistency may no longer be possible. A natural approach is to choose x so that its predicted measurements $\mathrm{H}x$ are close to y. This gives the *least-squares problem*

$$\boxed{\min_{x \in \mathbb{R}^N} \|y - \mathrm{H}x\|_2^2.} \tag{14.14}$$

The vector $y - \mathrm{H}x$ is called the *residual*. Its squared Euclidean norm is

$$\|y - \mathrm{H}x\|_2^2 = \sum_{m=0}^{M-1} \big|y[m] - (\mathrm{H}x)[m]\big|^2.$$

Thus the objective measures the total squared discrepancy between the observed and predicted measurements. In this optimization problem, y and H are given, while x is the variable to be determined.

Least squares uses the observation model, but it does not explicitly incorporate prior knowledge about the signal. For example, we may know that x has a sparse wavelet representation. To include such information, introduce a penalty R and consider

$$\boxed{\min_{x \in \mathbb{R}^N} \left\{ \|y - \mathrm{H}x\|_2^2 + \lambda R(x) \right\}, \qquad \lambda > 0.} \tag{14.15}$$

This is *regularized least squares*. The penalty is chosen so that signals consistent with the prior information have smaller values of R. The parameter λ controls the tradeoff between fitting the observations and favoring this structure.

These formulations place signal recovery in the language of linear algebra and optimization. We will study their solutions, and the effect of particular choices of R, in the next lecture.

14.16 The finite DWT as an orthogonal matrix

The same vector notation applies to the DWT. Fix a periodic DWT on a real signal of length $N = 2^I$. Filtering, downsampling, and collecting the output coefficients are linear operations. Since the transform produces exactly N coefficients, it can be written as

$$\boxed{\mathrm{DWT}\{x\} = c = \mathrm{W}x, \qquad \mathrm{W} \in \mathbb{R}^{N\times N}.} \tag{14.16}$$

The vector c contains the coarsest approximation and all detail coefficients in the order given in (14.8).

For real orthogonal wavelet filters and the corresponding periodic implementation, W is an orthogonal matrix:

$$\boxed{\mathrm{W}^{\mathsf{T}}\mathrm{W} = \mathrm{W}\mathrm{W}^{\mathsf{T}} = \mathrm{I}_N.} \tag{14.17}$$

This is the finite-dimensional counterpart of representing a signal in an orthonormal basis. In particular, the inverse transform is $x = \mathrm{W}^{\mathsf{T}}c$.

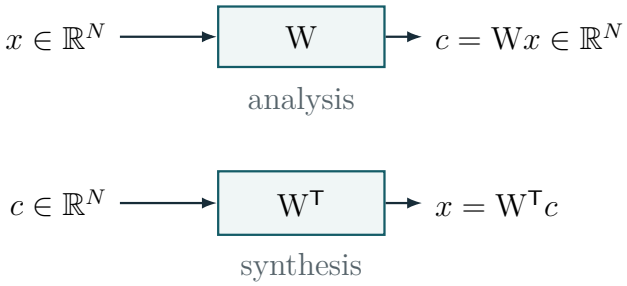


Figure 14.11: Analysis and synthesis for a real orthogonal periodic DWT. The analysis matrix is W, and the synthesis matrix is its transpose. Both maps are implemented by the filter-bank algorithm.

Orthogonality also gives the energy identity

$$\|\mathrm{W}x\|_2^2 = x^{\mathsf{T}}\mathrm{W}^{\mathsf{T}}\mathrm{W}x = x^{\mathsf{T}}x = \|x\|_2^2. \tag{14.18}$$

Writing the transform as a matrix is useful for analysis, but does not require us to form or store that matrix. The filter bank computes the same map in $O(LN)$ operations. For a biorthogonal DWT, the matrix is still invertible, but its inverse need not be its transpose and the energy identity need not hold.